Migration lifecycle - high-level flow

How a migration is created and used. Focus: Introspect Schemas, Validate Plan, and Materialize Flows - when they should or must run, and in what order.

1. End-to-end path (happy path)

Recommended sequence for a new migration.

```
1. Create migration project
2. Setup: source + destination connections (and system ids)
3. Plan: stages via Generate with AI and/or Add extraction / mapping
4. Edit mappings (Plan editor or Mapping tab)
5. Introspect Schemas   (SHOULD - MySQL/Postgres)
6. Validate Plan        (SHOULD - before materialize)
7. Materialize Flows    (MUST before Run)
8. Run: Sample / Pilot / Full
```

2. What each Plan action does

Introspect Schemas - SHOULD (strongly recommended for DB sources)
Reads live column lists from source and destination connections (MySQL/Postgres) using each stage's real table names (sourceEntity / destinationEntity). Stores them on plan_config.schemas. Feeds mapping Known fields and Validate (required NOT NULL columns, types). Not required for materialize/run to succeed. Not available for non-SQL runners. Re-run after you add or change stage table names.

Validate Plan - SHOULD (Materialize re-checks and can block)
Checks stages and mappings: extract has sourceEntity, transform has destination, required destination columns covered, types/lookups sensible. Saves lastValidation on the project. Does not create Data Flows or write data. Materialize runs the same validator again for plan v2 and BLOCKS if invalid - so Validate first is the practical path to a clean materialize.

Materialize Flows - MUST before Sample / Pilot / Full
Creates or updates Data Flows from stages; links migration_project_id, migration_stage_id, and entity_key. Extract -> retrieve/ingest. Transform/mapping -> dual-sink (mapped ingest + destination export) wired to the extract with the same entity_key. Needs source connection; maps need destination. Re-run after stage or mapping changes. Without it, Run has nothing executable.

3. Must / should / optional

Action	When	Blocking?
Setup connections	Before extract/map that hit DBs	Materialize needs source (+ dest)
Plan stages	Before introspect/validate/materialize	Yes - need stages
Introspect Schemas	After table names exist	No - improves UI + Validate
Validate Plan	After stages+mappings	Materialize fails if invalid (v2)
Materialize Flows	After plan is valid; before Run	Yes - Run needs flows
Run sample/pilot/full	After Materialize	Uses linked Data Flows

4. Sequence (Plan quality gates)

```
Stages ready (AI and/or manual)
            |
            v
   [ Introspect Schemas ] ---- optional but recommended
            |                    (schemas -> UI + Validate)
            v
   [ Validate Plan ] --------- recommended; Materialize
            |                    re-checks and may refuse (v2)
            v
   [ Materialize Flows ] ----- required: create/update Data Flows
            |
            v
   [ Run Sample / Pilot / Full ]
```

5. How pieces link (AI and manual are the same)

AI Generate and manual Add extraction / Add mapping write the same migration_stages rows. Shared keys: entity_key (pairs extract with map), config.sourceEntity / destinationEntity (real tables), config.fieldMappings. Materialize names flows Extract {entity} / Map {entity} and stores migration_project_id and migration_stage_id on each Data Flow. Run walks stages in sort_order and executes each linked flow.

6. Practical checklist

1. Create migration; set source + destination connections on Setup.
2. Plan: Generate with AI and/or Add extraction then Add mapping (same entity_key).
3. Set real Source table / Destination table; edit field mappings.
4. Introspect Schemas (if SQL) to refresh Known fields.
5. Validate Plan; expand result; fix errors.
6. Materialize Flows; Open flow links appear.
7. Run Sample, then Pilot, then Full as needed.
8. After mapping/stage edits: Validate (if unsure) -> Materialize again -> re-run.

Note: Clear ingest / clear run history are Run-tab tools, not plan creation. Introspect does not change stages. Validate does not create flows. Materialize does not migrate data by itself - Run does.